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Executive Summary

PropOS is a comprehensive, cloud-native SaaS platform designed to digitise and optimise the full lifecycle of real estate, property management, and facility operations across the GCC. It serves eight distinct stakeholder roles — from field technicians to C-level executives — through seven purpose-built portals delivered on web and mobile. Built for the Saudi Vision 2030 real estate boom, PropOS is ZATCA Phase 2 compliant, Arabic-first, and deployable on AWS Riyadh region with full data residency.

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SECTION 1

Platform Vision & Positioning

PropOS is the **operating system for real estate** — a multi-sided, cloud-native SaaS platform that unifies property management, facility operations, enterprise accounting, and investor intelligence under one roof. Positioned for the GCC market with Saudi Arabia as the primary launch territory, PropOS capitalises on Vision 2030's real estate expansion and the growing demand for digital-first property operations.

Strategic Pillars

Digital-First Operations End-to-end automation of property and facility workflows, eliminating paper and manual processes	Investor Intelligence Real-time portfolio analytics, feasibility studies, and market benchmarking for property owners	Workforce Empowerment Mobile-first tools that enable field staff to deliver professional, measurable service
Regulatory Compliance ZATCA Phase 2, PDPL, NCA cybersecurity, and Ejar integration built-in from day one	Scalable Architecture Cloud-native, multi-tenant, microservices design that scales from 10 to 10,000 clients	Arabic-First Experience RTL design, Arabic typography, and localised UX — not an afterthought but a foundation

Market Opportunity

Primary Market	Saudi Arabia — 1.3M+ residential units, SAR 500B+ real estate sector, Vision 2030 NEOM & giga-projects driving demand
Secondary Markets	UAE, Kuwait, Bahrain, Qatar, Oman — similar regulatory frameworks and Arabic language requirement
Target Segments	Property management companies, real estate developers, institutional investors, family offices, government REITs
Competitive Gap	No dominant Arabic-first, ZATCA-compliant, full-stack property & facility management SaaS currently exists in KSA
Revenue Potential	TAM: SAR 2.4B SAM: SAR 480M SOM (Year 3): SAR 24M ARR

SECTION 2

User Personas & Roles

PropOS serves eight distinct stakeholder types, each with purpose-built interfaces, data access levels, and workflow tools. Role-based access control (RBAC) ensures every user sees exactly what they need — no more, no less.

Role	Primary Goals	Key Portals	Critical Features
Super Admin	Platform governance, multi-tenant management	Super Admin Panel	Tenant mgmt, feature flags, platform analytics, billing
Admin (Company)	User & role management, company config	Admin Portal	RBAC editor, integrations, audit logs, subscription mgmt
Executive (C-Level)	Strategic oversight, performance monitoring	Executive Dashboard	KPI/OKR tracker, P&L, portfolio performance, board reports
Property Manager	Day-to-day property operations	Manager Portal	Lease pipeline, tenant comms, maintenance approvals, budget
Facility Manager	Asset & maintenance management	Manager Portal	Preventive maintenance, asset register, vendor management
Finance Manager	Full accounting operations	Finance Portal	GL, AR/AP, ZATCA invoicing, VAT, payroll, reconciliation
Employee / Technician	Execute work orders & inspections	Employee Portal + Mobile App	Work orders, GPS check-in, inspection checklists, knowledge base
Customer (Owner/Investor)	Monitor portfolio & financial performance	Customer Portal	NOI reports, feasibility studies, occupancy, payment tracking
Tenant	Self-service rent & maintenance	Tenant Portal	Online rent payment, maintenance requests, lease documents

SECTION 3

Seven Platform Portals

3.1 Public Marketing Website

Purpose	Convert visitors into leads and paying subscribers; showcase brand credibility
Key Sections	Hero with ROI calculator, Services overview, How It Works, Pricing plans, Case studies, Blog/SEO hub, Demo request, About Us, Careers
Unique Features	Interactive ROI calculator (enter property value → see projected yield), rental demand heat map by district, bilingual AR/EN with RTL toggle
Tech	Next.js 14 (SSR + SSG for SEO), TailwindCSS, Framer Motion animations, Crisp chat widget

3.2 Customer Portal — Property Owners & Investors

Dashboard	Portfolio valuation widget, occupancy gauge, monthly revenue vs prior period, maintenance alerts, lease renewal pipeline
Financial Intelligence	NOI tracker, cash flow waterfall, revenue vs expense (12-month), VAT reports, owner distribution statements, comparative yield vs market
Feasibility Studies	AI-generated ROI projections, cap rate, payback period, comparable sales data, rental demand index, risk scoring, branded PDF export
Subscription Tiers	Starter (5 units) / Professional (25 units + feasibility) / Investor (unlimited + AI insights + API)
Additional Modules	Tenant profiles & risk scoring, real-time work order statuses, payment centre, document vault, quarterly market reports

3.3 Employee Portal — Field Staff & Technicians

Work Orders	P1–P4 priority levels with SLA timers, GPS check-in verification, photo/video evidence, parts logging, tenant signature capture, auto-escalation
Inspections	Move-in/out checklists, condition rating + photos, common area rounds, fire safety compliance, auto-generated PDF reports, punch-list → work order linking
Mobile-First	Offline capability, QR asset scanning, voice notes, dark mode for low-light environments
Personal Dashboard	My work orders, SLA score, customer ratings, training certifications, timesheet & attendance
Knowledge Base	Searchable SOPs, equipment manuals, how-to videos — available offline

3.4 Executive Dashboard

Command Centre	Real-time portfolio value, occupancy, MRR/ARR trend, P&L; snapshot, CSAT/NPS, workforce utilisation heatmap
OKR & KPI Framework	Company OKRs with cascading to teams/individuals, 40+ pre-built real estate KPIs, Red/Amber/Green status, trend arrows, auto-updated from live data
Predictive Analytics	6-month occupancy forecast, rent arrears risk scoring, maintenance cost forecast by asset age, market opportunity alerts
Board Reporting	One-click PDF executive summaries, investor-ready portfolio performance matrix, risk register

3.5 Manager & Operations Portal

Property Management	Occupancy pipeline, lease renewal queue, tenant communication log, maintenance approvals, vendor POs, budget vs actual
Facility Management	Asset register (HVAC, elevators, generators, MEP), preventive maintenance calendar, contractor scheduling, energy consumption monitoring
Workforce Tools	Drag-and-drop employee assignment, skills matrix, workload balancing, contractor management, vendor performance scoring
Approval Workflows	Configurable multi-level approval chains by spend threshold — field → manager → finance → CEO

3.6 Finance & Accounting Portal

Core Accounting	Multi-entity GL, journal entries, cost centres per property/unit/department, AR/AP, bank reconciliation, multi-currency (SAR, USD, AED, KWD)
ZATCA Compliance	Phase 2 e-invoicing (Fatoorah API), QR-coded invoices, VAT return preparation (15% KSA), debit/credit note management
Payroll	Salary processing, GOSI contributions, end-of-service provisions, payslip generation, WPS compliance, Mudad/Qiwa integration
Budgeting	Annual budget per property/department, monthly variance analysis, rolling 12-month forecast, scenario modelling (optimistic/base/pessimistic)
Financial Statements	P&L, Balance Sheet, Cash Flow Statement — consolidated and per-entity — export to Excel/PDF

3.7 Admin & Super Admin

Admin	User management (invite, deactivate, reset MFA), RBAC role builder, company branding, integration config, audit log viewer, subscription & billing
Super Admin	Multi-tenant company registry, platform-wide analytics, feature flags, subscription revenue MRR dashboard, system health, white-label settings
White-Label Engine	Custom domain, logo, color scheme, email templates — premium B2B tier for large property management firms

SECTION 4

Enterprise Feature Modules

Community Management Owner associations, voting, shared cost allocation, notice board, visitor management, amenity booking for residential compounds	Smart Building / IoT BMS integration (Siemens, Honeywell, Schneider), real-time energy monitoring, sensor fault alerts, smart access control, meter automation	AI Lease Abstraction Upload PDF lease → AI extracts key terms, dates, obligations, renewal clauses, and auto-populates all fields
Digital Document Hub Draft → Review → e-Sign (DocuSign/Verified.sa) → Store → Remind on expiry — full version history	Tenant Self-Service Online rent via Mada/Apple Pay/SADAD, maintenance requests with photos, lease access, receipt downloads, manager communication, service rating	Visitor & Access Mgmt Front-desk check-in, pre-registered visitors, QR visitor passes, intercom integration, parking management
ESG / Sustainability Carbon footprint per building, energy efficiency scoring, water usage tracking, ESG reporting for institutional investors	Vendor Marketplace Platform-approved contractor directory, bidding system for large jobs, vendor rating & SLA performance, invoice automation	Training & Compliance Employee training records, certification expiry alerts (fire warden, first aid), mandatory course tracking, LMS integration
Emergency Response Incident reporting, emergency contact tree, evacuation checklists, post-incident PDF report, regulatory notification	Multi-Entity Accounting Consolidated financials across multiple legal entities, inter-company transactions, entity-level P&L; and balance sheet	Predictive Maintenance ML models trained on asset age, usage patterns, and historical failures — predicts next failure with 80%+ accuracy
Lease Renewal Engine Auto-identifies leases expiring in 90/60/30 days, generates renewal proposals, tracks negotiation status, sends reminders	Utility Management Electricity, water, gas consumption per unit, automated billing, consumption anomaly detection, tenant utility statements	Insurance Management Building and unit insurance policy tracking, premium schedules, claims management, renewal alerts, certificate storage
Portfolio Analytics Cross-property performance comparison, neighbourhood benchmarking, cap rate analysis, investment return optimisation	Communication Centre Internal team chat (per building), tenant broadcast messaging, WhatsApp Business integration, email campaign manager	Mobile Field App GPS work orders, offline inspections, QR asset scanner, biometric login, photo annotation, voice notes, push notifications

SECTION 5

Subscription & Business Model

Customer Tiers (B2C — Property Owners)

Plan	Price (SAR/mo)	Units	Key Features
Starter	199	Up to 5	Basic financial reports, maintenance tracking, tenant management, document vault
Professional	599	Up to 25	Full dashboards, feasibility studies, market reports, ZATCA invoicing, mobile app
Investor	1,499	Unlimited	AI insights, predictive analytics, custom reports, API access, priority support, white-glove onboarding

Operations Tiers (B2B — Property Management Companies)

Plan	Price (SAR/mo)	Employees / Units	Key Features
Growth	1,499	15 / 100	Core property & facility mgmt, employee portal, basic reporting, mobile app
Business	3,999	50 / 500	Full accounting, ZATCA, payroll, advanced analytics, vendor marketplace, SLA management
Enterprise	Custom	Unlimited	White-label, dedicated CSM, SLA guarantee (99.9%), SAP/Oracle connectors, on-premises option, custom integrations

Additional Revenue Streams

Tenant Portal Add-On	SAR 15/unit/month for tenant self-service portal activation
On-Demand Feasibility Study	SAR 499 per report (included in Professional+ subscriptions)
Vendor Marketplace Commission	5–10% commission on contracts facilitated through the platform
Implementation Services	One-time onboarding: Growth SAR 4,000 Business SAR 15,000 Enterprise SAR 50,000+
Training & Certification	SAR 299–999 per course per employee (fire safety, facilities management, etc.)
White-Label Licensing	SAR 15,000+/month — platform under client's brand with custom domain
Data Insights API	SAR 999/month for third-party BI tool access (Power BI, Tableau connectors)

SECTION 6

Technology Stack

Frontend Web	Next.js 14 (App Router + SSR), TypeScript, TailwindCSS, shadcn/ui, TanStack Query, Zustand, i18next (AR/EN RTL), ApexCharts, Mapbox GL JS, Framer Motion
Mobile Apps	React Native (Expo) — iOS & Android from single codebase. Offline SQLite sync, GPS/geofencing, camera + annotation, biometric auth, QR scanner, push notifications (FCM + APNs)
Backend Framework	NestJS (Node.js + TypeScript) — modular monolith evolving to microservices. Prisma ORM, Passport.js auth, Bull queue (Redis-backed), Socket.io WebSockets
APIs	REST (public client-facing), GraphQL (dashboard data queries — avoid over-fetching), gRPC (internal service-to-service), WebSockets (real-time notifications & chat)
Primary Database	PostgreSQL 15 — multi-tenant with Row-Level Security (RLS). Every table has tenant_id enforced by Postgres RLS policies
Caching & Queue	Redis (ElastiCache) — sessions, API caching, rate limiting, pub/sub, Bull job queues for async workflows
Search & Analytics	OpenSearch (AWS) — full-text search across all entities, analytics aggregations for dashboards
Time-Series (Phase 3)	TimescaleDB — IoT sensor data, energy consumption time-series, predictive maintenance datasets
File Storage	AWS S3 — documents, images, inspection photos, reports. CloudFront CDN for delivery. Pre-signed URLs for secure access
Message Queue	AWS SQS + SNS — event-driven workflows (invoice → ZATCA → notification → ledger update)
AI / ML (Phase 3)	Python FastAPI microservice with scikit-learn / TensorFlow: predictive maintenance, tenant churn, rental price optimisation, lease abstraction (NLP)
DevOps	GitHub Actions (CI), ArgoCD (GitOps CD), Terraform (IaC), Helm (Kubernetes charts), Datadog (observability), Sentry (error tracking)

SECTION 7

Technical Architecture

PropOS adopts a **Modular Monolith** → **Microservices** evolution strategy. The initial version is a well-structured NestJS monolith with clear module boundaries — faster to build, easier to debug, and deployable on a single Kubernetes deployment. As traffic and team size grow, modules are extracted into independent services without rewriting the application.

Core Services (when matured)

Service	Responsibility	Key Technologies
auth-service	JWT, MFA, SSO/SAML, OAuth2, RBAC, API keys	NestJS, Redis, PostgreSQL
property-service	Buildings, units, leases, tenants, contracts	NestJS, PostgreSQL, S3
maintenance-service	Work orders, inspections, assets, PM schedules	NestJS, PostgreSQL, Redis pub/sub
accounting-service	GL, AR, AP, ZATCA e-invoicing, bank reconciliation	NestJS, PostgreSQL, ZATCA API
analytics-service	KPI aggregation, dashboards, ML model serving	Python FastAPI, OpenSearch, Redis
notification-service	Email, SMS, WhatsApp, push notifications	NestJS, SQS, Firebase, AWS SES
document-service	File storage, e-signing, PDF generation, OCR	NestJS, S3, DocuSign API, Python
workflow-service	Approval chains, automation rules, triggers	NestJS, Bull, PostgreSQL
iot-service (Phase 3)	BMS integration, sensor data ingestion, alerts	Python, MQTT, TimescaleDB

Multi-Tenancy Strategy

Approach	Row-Level Security (RLS) in PostgreSQL — single shared database with tenant isolation enforced at DB level via Postgres RLS policies on every table
Schema Isolation	Optional schema-per-tenant for large Enterprise clients requiring stronger data isolation (e.g., government entities, large REITs)
Tenant Identification	JWT token includes tenant_id claim → NestJS middleware injects into every DB query → Postgres RLS policy validates automatically

Cross-Tenant Queries	Super Admin service uses a privileged connection that bypasses RLS for platform-wide analytics and health monitoring
Data Residency	All tenant data stored in AWS Riyadh region (me-south-1) by default. UAE clients can optionally use AWS UAE (me-central-1)

SECTION 8

Mobile Application Strategy

Framework	React Native (Expo) — single TypeScript codebase compiles to native iOS and Android apps. Shares 85%+ of business logic with web frontend
Target Users	Primary: Field Employees & Technicians (daily active users). Secondary: Managers (approvals & monitoring). Tertiary: Property Owners (portfolio checks)
Offline-First	SQLite local DB via WatermelonDB — work orders, inspection checklists, and asset data available without connectivity. Auto-sync when back online
GPS & Geofencing	Mandatory GPS check-in/out for work orders with geofence validation (must be within 100m of property). Background location tracking for field managers
Camera & Media	Photo annotation (draw arrows, add text overlays on inspection images), video capture for major defects, automatic compression and S3 upload
Biometric Auth	Face ID / Touch ID login. Fallback to PIN. JWT refresh tokens stored in iOS Keychain / Android Keystore — not AsyncStorage
QR & Barcode Scanner	Scan asset QR codes to instantly access maintenance history, manuals, warranty info, and create work orders. Generate QR codes for new assets
Push Notifications	FCM (Android) + APNs (iOS) via Firebase. Real-time alerts: new work order, SLA breach, manager approval, payment received, lease expiry
Arabic RTL Mobile	Full RTL layout support, Arabic keyboard, Arabic numerals (optional), bilingual push notification content based on user language preference
App Store Strategy	Separate branded apps: 'PropOS Field' (employees), 'PropOS Owner' (customers). Enterprise white-label versions available under client brand

SECTION 9

Cloud Deployment — Zero to Hero

PropOS is deployed on **AWS (preferred — Riyadh region me-south-1)** with full IaC via Terraform, GitOps via ArgoCD, and container orchestration via Amazon EKS. The deployment follows a phased approach from foundation to full production-scale infrastructure.

AWS Organization + Control Tower setup, IAM roles & SCPs, VPC with public/private subnets, NAT gateways, Terraform state backend (S3 + DynamoDB lock), GitHub Organization + Turborepo monorepo, GitHub Actions CI pipeline, AWS Secrets Manager, Route 53 DNS, ACM SSL certificates, basic CloudWatch alarms & SNS alerting

Amazon EKS cluster (production + staging namespaces), RDS PostgreSQL Multi-AZ (db.r6g.xlarge primary + read replica), ElastiCache Redis cluster (cache.r6g.large), S3 buckets with lifecycle policies + versioning, CloudFront CDN (static assets, media, public website), AWS SES (transactional email), OpenSearch domain (analytics + search), ECR container registry, SQS/SNS message queues

Kubernetes namespaces per service, Helm charts for each microservice, HPA (Horizontal Pod Autoscaler) configs, ALB Ingress controller with path-based routing, ArgoCD GitOps continuous deployment, Secrets injection (AWS Secrets Manager → K8s secrets via External Secrets Operator), Blue/green deployment with automated rollback, Flyway/Prisma zero-downtime DB migrations, k6 load testing before each major release, WAF rules (OWASP Top 10 protection)

Full observability: Datadog (APM, infrastructure, logs), Sentry (error tracking + source maps), OpenTelemetry distributed tracing across all services, Pingdom uptime monitoring, automated daily RDS snapshots (30-day retention + cross-region copy to UAE), DR runbook tested quarterly (RTO: 4h, RPO: 1h), AWS Shield Standard (DDoS baseline), annual penetration testing by certified third party

Read replicas for analytics query offloading, RDS Proxy for connection pooling, ElastiCache cache-aside pattern for heavy dashboard queries, Reserved Instances + Savings Plans (40–60% compute cost reduction), multi-region active/passive (primary: Riyadh, failover: UAE), AWS WAF + Shield Advanced, CDN edge optimisation for GCC PoPs

Estimated Monthly Cloud Cost

Scale	Active Tenants	Est. Monthly AWS Cost	Est. MRR	Cloud % of Revenue
MVP	50–100	SAR 7,500–15,000	SAR 100K–300K	5–7%
Growth	500–1,000	SAR 30,000–55,000	SAR 750K–1.5M	3–4%
Scale	3,000–5,000	SAR 100,000–180,000	SAR 4.5M–7.5M	2–3%
Enterprise	10,000+	SAR 300,000+	SAR 15M+	<2%

SECTION 10

UI / UX Strategy

Design Philosophy	Data clarity over decoration. Progressive disclosure — summaries first, details on demand. Never show a user data that isn't actionable
Arabic-First Design	All layouts built RTL from day one. Arabic typography: IBM Plex Arabic (interfaces) + Noto Naskh Arabic (documents). RTL charts, tables, nav flows
Design System	Proprietary component library in Figma + Storybook. Design tokens (colors, spacing, radius, shadows, typography scale) established before any component
Role-Based Onboarding	First-login experience tailored per role: technician → work order quickstart; executive → KPI setup; owner → portfolio wizard. No generic onboarding
Customisable Dashboards	Drag-and-drop widget system for Executive and Manager portals. Users pin their KPIs. Saves cognitive load and increases daily active usage
Accessibility	WCAG 2.1 AA compliance. Minimum 4.5:1 contrast ratios. Full keyboard navigation. Screen reader support (NVDA, VoiceOver). ARIA labels throughout
Dark Mode	System-aware dark mode for all portals. Critical for field staff using mobile in low-light environments (basements, plant rooms, night shifts)
Performance	LCP < 2.5s, FID < 100ms, CLS < 0.1 (Core Web Vitals targets). Skeleton loaders — never blank screens. Optimistic UI updates for instant feel
Mobile UX Principles	Thumb-zone design (critical actions in bottom 60% of screen). Minimum 44×44pt tap targets. One-tap work order status updates. Voice notes
Micro-interactions	Status change animations, progress rings on SLA timers, toast notifications with undo, contextual empty states with actionable CTAs, celebration moments

SECTION 11

Security & Compliance

Authentication	OAuth2 + JWT (15-min access / 7-day refresh tokens). MFA mandatory for Finance and Executive roles (TOTP via Authenticator app + SMS fallback). SSO via SAML 2.0 for enterprise clients
Authorisation	RBAC with least-privilege principles. Field-level permission control for sensitive financial data. Custom roles builder for each company. Permission audit log
Data Protection	AES-256 encryption at rest (RDS + S3). TLS 1.3 in transit. End-to-end encryption for sensitive documents. Data residency: AWS Riyadh (me-south-1)
ZATCA Compliance	Phase 2 Fatoorah API integration, QR-coded e-invoices, VAT return automation, debit/credit note management — all GAZT-approved workflows
PDPL Compliance	Saudi Personal Data Protection Law — data classification, consent management, right to deletion workflows, data processing agreements with vendors
NCA Compliance	National Cybersecurity Authority Essential Controls — asset inventory, patch management, incident response, privileged access management
Audit Trail	Every data mutation logged: user, timestamp, IP address, HTTP method, entity, before/after values. Immutable append-only audit log in separate DB table
Penetration Testing	Annual external pen test by certified third party (CREST-accredited). Quarterly internal vulnerability scans. Bug bounty programme (Phase 2)
Certification Targets	ISO 27001 (Year 2), SOC 2 Type II (Year 2), PCI DSS Level 3 (if processing cards directly)
Incident Response	Defined runbooks for: data breach (notify NCA within 72h), DDoS, credential compromise, ransomware. Annual tabletop exercises. SIEM with Datadog

SECTION 12

Integrations Map

Category	Integration	Purpose
Saudi Government	ZATCA Fatoorah API	Phase 2 e-invoicing, VAT returns, QR-coded invoices
Saudi Government	Ejar Platform	Rental contract registration with Ministry of Housing
Saudi Government	Absher API	National ID verification for tenant onboarding (KYC)
Saudi Government	Wathq	Commercial registry verification for company onboarding
Saudi Government	Mudad / Qiwa	WPS payroll compliance, labour contract registration
Saudi Government	Aqar Registry	Real estate title deed verification and registration
Payments	HyperPay	Card processing (Visa/Mastercard/Amex) — PCI DSS compliant
Payments	Mada / STC Pay	Local Saudi debit & mobile wallet processing
Payments	SADAD	Saudi bill payment network — automatic rent collection
Communication	WhatsApp Business API	Tenant notifications, lease reminders, payment confirmations
Communication	Unifonic / Twilio	SMS OTP, alerts, broadcast notifications in KSA
Communication	Firebase + APNs	Mobile push notifications (Android + iOS)
ERP	SAP S/4HANA	GL sync, PO management, consolidated financials (Phase 3)
ERP	Microsoft Dynamics 365	CRM sync, finance module integration (Phase 3)
Digital Signature	DocuSign / Verified.sa	Legally binding e-signatures on leases and contracts
Maps & GIS	Google Maps Platform + Mapbox	Property locations, field staff routing, district analytics
IoT / BMS	Siemens Desigo / Honeywell	Building automation, HVAC, energy monitoring via MQTT/REST
Analytics	Power BI / Tableau	Enterprise BI export via PropOS Data API connector
Identity	SAML 2.0 / Azure AD	SSO for enterprise clients using Microsoft 365

SECTION 13

KPIs, OKRs & Executive Metrics

Real Estate Operations KPIs

KPI	Formula / Definition	Target Benchmark
Occupancy Rate	Occupied Units / Total Units × 100	≥ 92%
Net Operating Income (NOI)	Gross Revenue – Operating Expenses (excl. debt)	Positive YoY growth
Gross Rental Yield	Annual Rent / Property Value × 100	≥ 6% (KSA market)
Cap Rate	NOI / Current Property Value × 100	Market-dependent
Rent Collection Rate	Rent Collected / Rent Billed × 100	≥ 97%
Tenant Turnover Rate	Vacated Units / Avg. Occupied Units × 100	≤ 15% annually
Average Days to Lease	Days from vacancy to signed lease	≤ 21 days
Maintenance Cost / Unit	Total Maintenance Cost / Total Units (monthly)	Benchmark vs prior year
Lease Renewal Rate	Renewed Leases / Expiring Leases × 100	≥ 70%

Facility Management KPIs

KPI	Definition	Target
MTTR (Mean Time to Repair)	Avg time from work order creation to close	P1: ≤ 2h P2: ≤ 8h
SLA Compliance Rate	Work orders closed within SLA / Total work orders	≥ 95%
PM Compliance Rate	Preventive maintenance tasks completed on schedule	≥ 98%
Work Order Completion Rate	Closed work orders / Total work orders (monthly)	≥ 90%
Asset Utilisation Rate	Active operating time / Total scheduled time	≥ 85%
Energy Cost / sq/m	Total energy cost / Total managed area (monthly)	Reduce 5% YoY
First-Time Fix Rate	Work orders resolved without revisit / Total WOs	≥ 85%

SECTION 14

Implementation Roadmap

Phase 1 — Foundation & MVP	Months 1–6
Public marketing website (bilingual, SEO-optimised), authentication & RBAC, basic property/unit/tenant management, basic work order system (web), simple owner financial dashboard, core notification system, basic employee task view, Mada/HyperPay payment integration, ZATCA MVP e-invoicing. Goal: first 50 paying customers, validate core value proposition, SAR 200K ARR.	
Phase 2 — Core Platform	Months 7–12
Full accounting module (GL, AR, AP, bank reconciliation), customer portal subscription product, employee mobile app (iOS + Android), manager operations portal, preventive maintenance scheduler, document vault + e-signature, advanced reporting engine, tenant self-service portal, WhatsApp Business integration, SADAD bill payment. Goal: full operational replacement for mid-size property management firms, SAR 1.2M ARR.	
Phase 3 — Intelligence & Scale	Months 13–18
Executive dashboard & OKR management system, AI feasibility study reports, predictive maintenance ML models, IoT/BMS sensor integration, ESG sustainability module, vendor marketplace, white-label engine, Ejar/Absher/Wathq government API integrations, community management module, advanced AI analytics suite. Goal: enterprise accounts, regional market leadership, SAR 4M ARR.	
Phase 4 — Ecosystem & Scale	Months 19–24
Open API developer platform & marketplace, SAP/Oracle ERP connectors, multi-region deployment (UAE, Kuwait, Bahrain), AI lease abstraction (NLP), AI tenant chatbot (support + queries), BI connector (Power BI, Tableau), mobile offline sync improvements, IPO-readiness financial reporting, ISO 27001 + SOC 2 certification. Goal: regional SaaS leader, SAR 12M ARR, Series A fundraising.	

First 90 Days — Sprint Zero

Weeks 1–2	Finalise product scope, write detailed user stories for MVP, set up Linear/Jira backlog, assemble core team (1 PM, 2 BE, 2 FE, 1 Mobile, 1 UI/UX, 1 DevOps)
Weeks 3–4	Figma design system (Arabic + English), AWS infrastructure Terraform setup, monorepo scaffolding (Turborepo), CI/CD GitHub Actions pipeline, auth service
Weeks 5–8	Public website (Next.js, bilingual, SEO), property/unit/tenant CRUD, basic work order flow, owner dashboard skeleton, first internal demo

Weeks 9–12	HyperPay integration, basic financial reports, employee mobile app (React Native), ZATCA e-invoicing MVP, first 5 beta customers onboarded
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SECTION 15

Financial Model & Unit Economics

SaaS Unit Economics

Average Revenue Per Account (ARPA)	B2C: SAR 599/mo avg B2B: SAR 2,500/mo avg Blended: SAR 1,800/mo
Customer Acquisition Cost (CAC)	Direct sales: SAR 8,000 Digital/inbound: SAR 1,500 Blended target: SAR 3,500
LTV (Lifetime Value)	Avg churn 8%/year → $LTV = ARPA \times (1/churn) = SAR\ 1,800 \times 12.5 = SAR\ 270,000$
LTV:CAC Ratio	$SAR\ 270,000 / SAR\ 3,500 = 77x$ (target > 3x — extremely healthy SaaS unit economics)
Payback Period	$CAC / (ARPA \times Gross\ Margin) = SAR\ 3,500 / (SAR\ 1,800 \times 75\%) = 2.6\ months$
Gross Margin Target	75–80% at scale (SaaS benchmark). Year 1: 55% (high infra/headcount). Year 3: 75%
Burn Rate (Year 1)	SAR 250,000–350,000/month (team: 10–15 people + infrastructure + marketing)
Break-Even ARR	Approx. SAR 5M ARR assuming 65% gross margin and SAR 270K/mo operating costs at that scale

ARR Growth Projection

Period	Accounts	Avg MRR/Account	MRR	ARR
End of Year 1	80	SAR 1,500	SAR 120,000	SAR 1.44M
End of Year 2	300	SAR 1,800	SAR 540,000	SAR 6.48M
End of Year 3	800	SAR 2,200	SAR 1,760,000	SAR 21.1M
End of Year 4	2,000	SAR 2,500	SAR 5,000,000	SAR 60M

Note: Projections assume 200% YoY growth in Years 1–2, 170% in Year 3, 185% in Year 4 — conservative by GCC SaaS standards for B2B property tech. Actual performance depends on go-to-market execution, team quality, and market conditions.

This